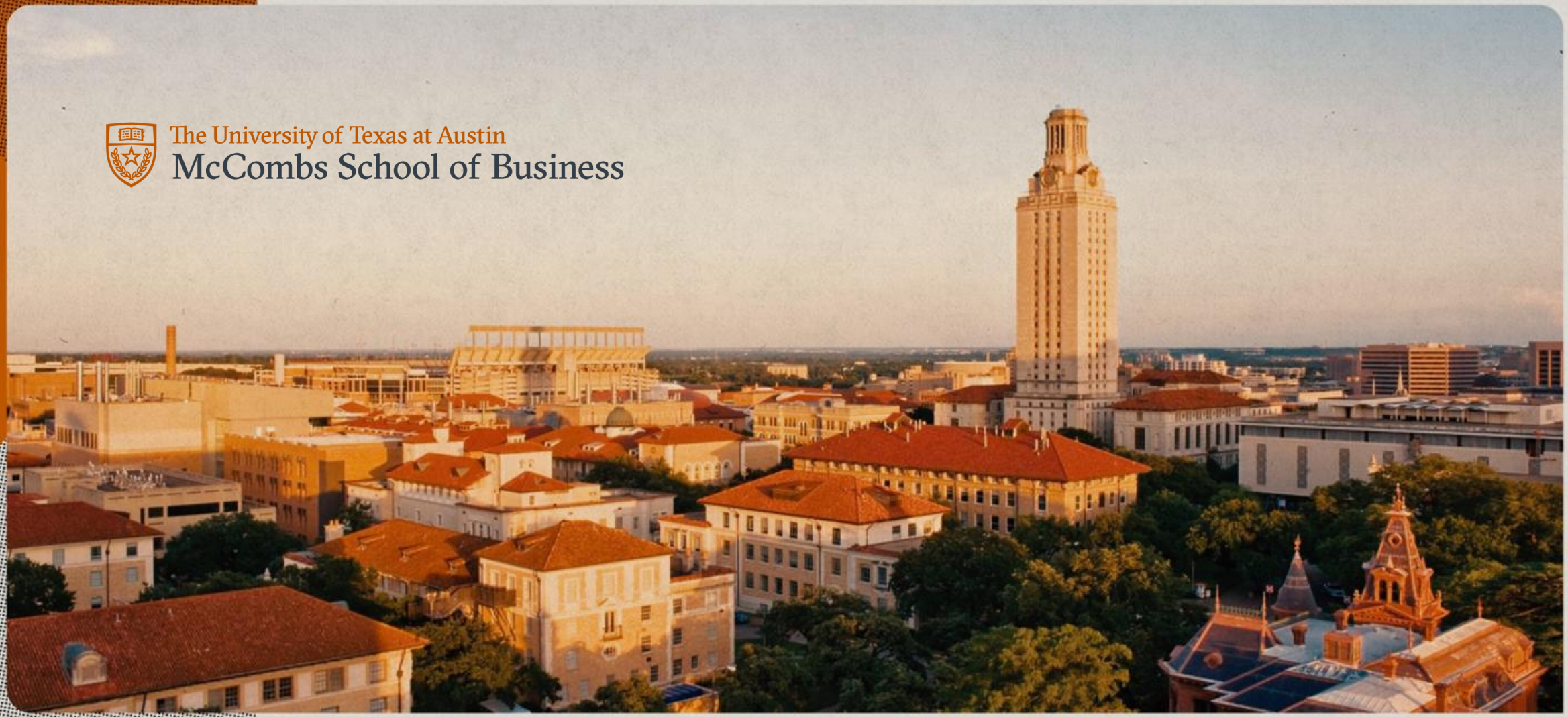

D6: Principal Components (Part 1)

Learning Objectives

1. Understand orthonormal transformations
2. Understand principal components transformation
3. How to interpret PCs
4. Relevance of linear algebra (eigenvectors, eigenvalues)

Major R commands

prcomp => solves for principal components

plot3d => visualization of principal components

Agenda

1. Discuss transformations, orthonormal transformations, principal components transformations
2. Principal components demo

   random100 (including Zone and defs).xlsx

3. Your turn

Transformations

In order to understand principal components, you should understand orthonormal transformations.

In order to understand orthonormal trnasformations, you should understand transformations.

Transformations

A **transformation** changes (transforms) one variable into another. Or more than one variable into others.

Ex. $y = x^2$ transforms x into y. This is a nonlinear transformation.

Ex. $y = 1 + 4x$ transforms x linearly into y.

Ex. $u = 4x + 1y$, $v = 3x - 4y$ transforms (x,y) linearly into (u,v)

Transformations

$u = 4x + 1y$, $v = 3x - 4y$ can also be written $\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 3 & -4 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

or $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, in which **A** = $\begin{bmatrix} 4 & 1 \\ 3 & -4 \end{bmatrix}$ is the **transformation matrix**.

We are concerned with **square** linear transformation matrices.

This **A** is square (2x2) and linear, but not orthonormal.

Transformations

An **orthogonal transformation** preserves angles.

An **orthonormal transformation** preserves angles and lengths.

$\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$ is **orthogonal** if the **inner product** of every pair of different columns in A is 0.

Orthogonal Transformations

Ex. $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, in which **A** = $\begin{bmatrix} 4 & 1 \\ 3 & -4 \end{bmatrix}$ is not orthogonal.

Ex. $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, in which **A** = $\begin{bmatrix} 4 & 3 \\ 3 & -4 \end{bmatrix}$ is orthogonal (but not orthonormal).

There are infinitely many orthogonal transformations of **x**.

Orthonormal Transformations

$\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$ is **orthonormal** if **A** is orthogonal and the length of each column of A is 1.

Ex. $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, in which **A** = $\begin{bmatrix} 0.8 & 0.6 \\ 0.6 & -0.8 \end{bmatrix}$ is orthonormal (but not principal component transformation.)

There are infinitely many orthonormal transformations of ***x***.

PC Transformations

$\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$ is the unique **principal component** transformation of ***x*** into ***u*** if the columns of **A** are the **eigenvectors** of the transformation of ***x*** into ***u***.

Ex. $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, in which **A** = $\begin{bmatrix} 0.707 & 0.707 \\ 0.707 & -0.707 \end{bmatrix}$ is the PC transformation of ***x*** into ***u***.

PC Transformations

- The PC transformation is orthonormal.
- The PC transformation is unique.
- The PC transformation maximizes the variance of the first component PC1; subject to that, maximizes the variance of the 2nd component PC2 (orthogonal to PC1); etc.
- The variance of component k is the kth eigenvalue.

Eigenvectors and Eigenvalues

The **eigenvectors** $\boldsymbol{x}$ of $\boldsymbol{u} = \boldsymbol{A}\boldsymbol{x}$, solve the determinantal equation $|\boldsymbol{A}\boldsymbol{x} - \lambda \boldsymbol{I}| = 0$. The **eigenvalues** are the solutions for $\boldsymbol{\lambda}$.

Fixed-point interpretation.

Geometric interpretation.

Variance interpretation.

Demo

PCA intro.R

Your turn

Continue the *PCA intro.R* demo with the *Stock data.xlsx* data.

- Run PCA using all three financial variables.
- Try to interpret the components.
- How many PCs would you retain?
- (If time) try 3d rotation of the original data for additional insights.